MPH — Automated Job Allocation Layer for the SMS — Project Report

Louta & Benguerir Mine Stockyards → MPH M16 Hub · M17 Sulphur — SP2M Program

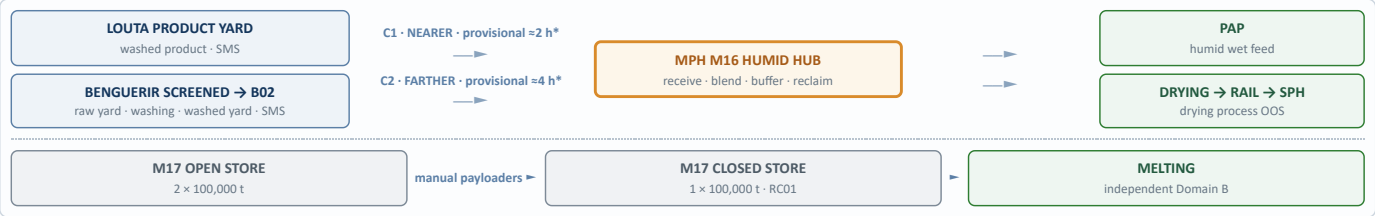
Team: Moncef Driouich | **Status:** Draft — Scope & Functional Design | **Rev.:** AC | **Date:** 8 September 2026

Rev.	Date	Scope of revision	Status
A-G	22 Jun-19 Jul 2026	Earlier scope, functional-design, interface, evidence and B02 development	Superseded - see source history
AA	28 Jul 2026	Job Schedule replaces the dispatch color/category with explicit non-material machine positioning: green for minute-reserved estimates and red for gaps requiring targeted replanning. Positioning carries time/from-to details but no tonnage; material allocation stays separate, and unresolved exact-profile jobs are withheld from ASP export until their timing validates	Draft — for analysis
AB	28 Jul 2026	Raw grades made mine-neutral and shared by Louta and Benguerir: RAW-TBC plus RAW-BT, RAW-BT-HIGH-CD and RAW-BT-LOW-CD. Site material profiles retain separate moisture, PSD and live-assay metadata; legacy grade IDs remain import-compatible	Draft — for analysis
AC	08 Sep 2026	Implemented weekly multi-site planning, dated multi-campaign demand, stage-aware physical/TSM mass conversion, persistent grade and yield configuration, consolidated timed resource controls, explicit results and M16 pile-construction blending with fixed or jointly optimised shares	Draft — for analysis

PART I — PROJECT SCOPE

1. Context

The phosphate planning boundary is the complete **three-site value chain**, not M16 alone. Louta, Benguerir and MPH are peer controlled yards in **Domain A**; each has an SMS and receives one coordinated **seven-day plan containing hourly jobs**. An optional rolling mode refines the first four hours to 15-minute decisions and retains an hourly tail; timed asset changes and issued jobs keep exact-minute boundaries. Mining and washing upstream of the stockyards are interfaces, not directly controlled plant processes; the S0 simulator nevertheless represents selected Louta and Benguerir campaigns as auditable process-transfer stages so that ore placement, reclaim, recovery, product stacking and corridor/train withdrawals conserve mass. **Domain B** plans M17 sulphur independently on the same platform.



*** Relative proximity is user-confirmed (project team, 22-Jul-2026).** The 30/60 km reassignment and resulting 2/4-hour lags are provisional modelling values; exact conveyor route lengths, effective speeds and transfer dwell remain TBC.

Site / stage	Planning role / output	Principal open evidence
Louta beneficiation stock route · SMS	110LG ore storage → modelled L1/L2 transfer → 130/140LG product reclaim → **C1 → M16**; 102MG remains reference-only	Full GA/reach and route matrix; live stocks/assays; 102MG→110LG route/rate; approved case-to-product campaign map
Benguerir six-pile chain · SMS	3 mine hoppers → 2 mine-ore piles → 2 screened/raw piles → 3 washing campaigns → 2 B02 product piles → **C2 → M16 / train**	Physical tags and GA positions; approved Benguerir line rates/recoveries, grade mapping and train orders
MPH M16 · SMS	Receive, blend and reclaim → PAP or drying	Final GA, route confluences and metric separation limits
MPH M17 · SMS	Independently plan closed-store feed → melting	Demand/calendars; open reclaim stays manual

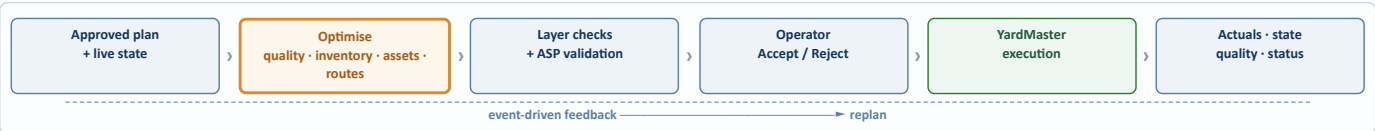
Phase 1 / expansion: M16 starts with PAP-1, 8 humid spaces and 3+2 machines. PAP-2 is already represented with the same demand template and the three explicit lifecycle states **Available / Disabled / Future**; its default is **Future**. Configuration supports N lines, 12 spaces and 4+3 machines. M17 closed storage expands from one to two stores and adds RC02. Mobile machines appear to be a separate package from the three solid-handling lots; final responsibility remains TBC.

System boundary: YardMaster/SMS maintains the 3D stock model and executes physical jobs. ASP supports manual authoring, rule validation and actuals tracking, but does not optimise demand, blends or multiple yards. This project is the **automated PlanAuthor above the three SMS**; ASP remains the vendor validation/execution bridge (§§14.2, 15.7–15.8).

2. Project Objective & Expected Impact

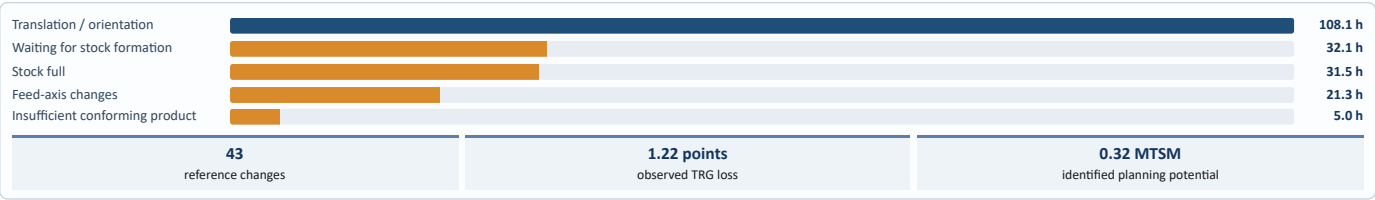
2.1 Objective

Turn approved dated demand and live yard state into an operator-supervised weekly rolling plan that coordinates quality, inventory, machines, conveyors, process lines, routes and corridor delays across Louta, Benguerir and M16; sulphur is solved separately. Each receiving unit may request several grade/quantity campaigns on the same day, with an intraday window and priority for each campaign.



2.2 Comparable OCP evidence — the planning loss already exists

The 2024 Khouribga Sidi-Chennane–Daoui diagnostic shows the same class of planning loss with only **2 stackers + 2 reclaimers**:



Comparable signal, not MPH forecast. Categories may overlap, Louta fleet data remain incomplete, and benefits must not be scaled linearly by machine count. Quantification requires de-overlapped replay, shadow mode and a supervised pilot.

2.3 Impact KPI scorecard

Operational targets remain **TBC** until an approved baseline, replay and pilot. Simulation may test **25% / 50% / 75%** recovery only on loss proven planning-addressable.

The Monitor makes these results explicit and shows the active objective weights beside the run. The current configurable objective includes unmet demand, shortage/safety exposure, movement, positioning, switching and quality deviation; the economic grade-cost term remains inactive until governed cost data exist. Weight values are scenario settings, not engineering facts, and the run stores them for reproducibility.

Area	KPI	Measure	Direction / fixed rule
Operations	Demand service	`Σ min(delivered,demand) / Σ demand`	↑ · TBC
	Feed interruption	demand > 0 and delivery = 0 [h]	↓ · TBC
	Corridor adherence	tonnes inside time/rate/quality window	↑ · TBC
Quality	On-spec mass	on-spec / delivered tonnes	↑ · TBC
	Off-spec exposure	off-spec tonnes + weighted deviation	↓ · TBC
	Give-away	weighted excess above target	↓ · TBC

Inventory/assets	Reconciliation drift	planning twin − SMS inventory	↓ · TBC
	Inventory bounds	time inside safety/capacity limits	↑ · overcapacity **0**
	Planning-addressable loss	de-overlapped hours + moves/starts per kt	↓ · 25/50/75% sensitivity
Planning	First-pass ASP	AUTO items with zero `ERROR` s	↑ · TBC
	Automation coverage	accepted AUTO / eligible	↑ · TBC
	Routine effort	operator authoring minutes / shift	↓ · TBC
Resilience	Replan latency	event → valid replacement	↓ · TBC
	Service recovery	disruption → restored feed	↓ · TBC
Governance	Schedule integrity	invalid/conflicting jobs	**0**
	Human-plan preservation	human items changed	**0 / 100%**
	Deterministic replay	same inputs/config/seed	**exact**
Integration	SMS ownership integrity	unmapped/wrong-SMS jobs	**0**

Claim boundary: baseline → offline replay → shadow mode → supervised pilot → sustained/economic validation. No throughput, financial, energy, staffing, reliability or safety benefit is claimed before pilot evidence. Formal acceptance measures are in §18.

3. Where this Layer Sits

Tier	Owner	Responsibility
Weekly / monthly planning	Operations/planning	Demand, campaigns, strategic stocks and expected supply; the application edits the approved seven-day demand envelope
Job Allocation Layer	**This project · L3**	Quality/quantity, blend, pile, machine, route and sequence
ASP + YardMaster/SMS	Vendor execution system	Validate, execute, maintain 3D state and return actuals
PLC / DCS / ACS	Control systems	Commands, interlocks, anti-collision and safety

Replanning is event-driven on quality drift, stock change, late supply, demand change, machine fault or execution deviation—not a fire-and-forget schedule.

4. Scope Boundary

In scope	Out of scope / retained authority
Coordinate a seven-day plan of hourly or mixed-resolution jobs across the 3 phosphate SMS; plan M17 independently	Generate the upstream monthly forecast or approve business demand
Select quality, quantity, blend, pile, machine, route and sequence	Mining, washing, PAP, drying and melting process operation
Direct target space when segregation requires; otherwise permit SMS auto-route	Physical actuation, deposition, 3D model and anti-collision
Track actuals, replan, explain and support what-if	Build/own SMS, SQMS, analysers, LIMS, CMS or equipment

Capability	Allocation layer	ASP	YardMaster / PLC / ACS
Multi-yard demand/blend optimisation	**Owns**	—	—
Physical route/shared-resource feasibility	**Owns** (§14.2)	Validates configured tuple	Executes route
Plan-item validation	Pre-checks	**Second gate**	—
Accept/reject/override	Preserves humans	Supports workflow	Executes accepted job
Safety/interlocks	Never replaces	—	**Retained authority**

An ASP "Route" is only `(Device, JobType, Location)`; it is not the conveyor network. RabbitMQ access/licensing and site master-data mappings remain TBC.

5. Functional Scope

Inputs	Decisions	Outputs
Dated demand campaigns: unit, grade, physical quantity, intraday window and priority	Source quality and quantity	Prioritised jobs per SMS/machine
SMS 1-m inventory, quality and age	Blend mode, component masses/shares and destination	Stack: source, grade, method, target pile + repère range, rate, tonnes
Actuals and machine calendars	Machine, pile, route and timing	Reclaim: pile + repère range, route, blend, rate, tonnes
Analyser + lab results	Reallocation on deviation	Profiles, alerts, rationale and KPIs

5.1 Repère position model

Each stockpile uses the operating scale **R0–R1000**, representing **1,000 equal longitudinal intervals**. A job may therefore state "store this grade on `LT-P1` from **R400 to R600**." Repères are operator-facing; physical metres remain canonical for SMS/ASP exchange, inventory segments, machine travel and safety distances:

$$x_m = x_0_m + (R / 1000) \times (x_1_m - x_0_m) \quad \cdot \quad R = 1000 \times (x_m - x_0_m) / (x_1_m - x_0_m)$$

The Innomotics FRS confirms numeric Start/End positions (`StartX` / `StopX`), per-metre pile inventory and position-range validation. The term **repère** and the R0–R1000 scale are a **project-team operating convention confirmed 24-Jul-2026**, not a claim that every pile is 1,000 m long. Before live use, each pile still needs its R0 endpoint and increasing direction, surveyed physical origin/usable length, and the ICD's StartX/StopX units and scale.

Constraint family	Planning rule
Corridors	C1/C2 capacity, availability, lag and in-transit inventory
Quality	Destination intervals, anti-contamination and stage-correct blending; dry-solids weighting for chemistry and physical-mass weighting for moisture
Inventory	Initial mass, safety stock, capacity and conservation
Machines	Reachability, exclusivity, rate, travel/setup, maintenance and rails
Same-pile work	M16 concurrency requires spatial/work-zone/route separation; limits TBC; mine applicability TBC
Sulphur	Open reclaim manual; closed RC01 and ≤8 start-stops/h planned
Dry stream	Dryer intake bounds humid reclaim; drying OOS

Detailed variables, constraints and equations are defined once in §§15.1–15.4.

6. Architecture & Interfaces

L4 · Business	ERP / production plan · Historian / reporting — managed OT/IT DMZ
L3 · Planning	**JOB ALLOCATION LAYER ↔ Operator HMI · LIMS · CMS**
L2 · Control	ASP / SMS ×3 · SQMS · PLC / DCS / ACS — jobs ▼ · actuals/state ▲
L1 · Field	Stackers · reclaimers · trippers · gates · analysers · belt scales

Counterpart	Dir.	Minimum exchange	Mechanism / cadence
SMS ×3	↔	Plan items; ASP issues; `JobUpdated` ; 1-m inventory/quality/age; machine state	Common YardMaster/RabbitMQ profile; per-site IDs TBC · event/continuous
SQMS / analysers	→	BPL, Cd, MgO, SiO_2, CaO/CO_2, moisture	OPC UA · continuous
LIMS	→	Detailed assay and PSD	API · per sample
ERP / planning	→	Approved demand, supply and campaigns	MES API/file · change
CMS	→	Availability and maintenance	API/OPC UA · event
Historian / HMI	↔	Trends, KPIs, plan, rationale and overrides	Historian + web/HMI · continuous

Deployment: dedicated L3 application on controlled OT; business exchanges cross a managed DMZ. Missing/stale SMS data are flagged with last-good-state plus operator fallback. **Configured:** sites, assets, reachability, route graph, specs, compatibility, patterns, weights, calendars, endpoints and IDs.

All equipment, conveyor and stockpile capacities and rates are represented in **physical/as-handled tonnes** at their material stage. TSM is a reported merchant-dry equivalent, not the capacity basis: raw physical mass converts through raw moisture and beneficiation yield; wet washed-product physical mass retains product moisture; after drying, 1 physical dry tonne = 1 TSM. The unified Grade configuration page maintains chemistry, yield/moisture, material profiles and receiving limits, and supports persistent add/delete with referential checks.

7. Representative Use Cases

Use case	Trigger	Decisions	Jobs	Replan
PAP on-grade humid feed	Dated campaign + pile quality	Build an eligible humid stockpile with fixed-matrix or optimized shares, then choose reclaim rate, machine and route	Component stacking followed by ready-boundary-safe reclaim	Cd/BPL, asset or demand drift
Drying / SPH	Spec + dryer limit	Build pile, humid blend, route	Stack/reclaim/gate	Quality, dryer or rail change
Sulphur continuity	Melting demand + closed stock	Replenishment and RC01 rate	Tripper/stack/closed reclaim	Delivery/payloader/RC01 deviation

8. Assumptions, Dependencies & Constraints

ID	Current ruling	Residual action / hedge
H1	Telegram access feasible in concept	Obtain ICD, credentials, licences, namespaces and keys; otherwise standard ASP client path
H2	**Resolved at solution level:** all sites use YardMaster	Configure three endpoints, IDs, routes, calendars and permissions
H3	Layer proposes; humans supervise; ASP arbitrates	`AUTO-` ownership only; human items fixed; audit retained
Data	Inputs assumed timely enough	Freshness limits, last-good fallback and alarms
Demand	Approved input-side demand authoritative	Validate source, cadence and governance

9. Deliverables & Phasing

Phase	Outcome	Exit evidence
0 · Scope/data	Boundaries, evidence and interfaces	Approved scope + gap owners
1 · Functional design	Twin, optimisation, constraints and ICDs	Design review + acceptance set
2 · Twin/shadow	Replay, simulation and supervised shadow	Baseline, validation and operator sign-off
3 · Deployment	Three-SMS closed loop and KPIs	Supervised pilot then acceptance

10. Key Risks

Risk	Control / proof
Stale/inaccurate quality or inventory	Freshness, reconciliation, fallback and actuals replay
Missing SMS ICD/access/master data	Vendor workshop, emulator and per-site mapping tests
Unreliable demand/route model	Input governance, route audit and zero-invalid-route acceptance
Asset/layout mismatch	GA/equipment list, reachability review and shadow validation
Operator distrust/conflicting edits	Rationale, Accept/Reject, human-item preservation and audit
Overstated benefits	De-overlap, approved baseline and no pre-pilot extrapolation

11. Success Criteria

The canonical outcome scorecard is §2.3; technical acceptance is §18.

Historical baseline › **Offline replay** › **Shadow mode** › **Supervised pilot**

12. Non-Functional Requirements

Requirement	Design intent	Status
Latency	Event-to-valid-replan fast enough to avoid avoidable gaps	TBC
Availability	Redundancy; last-good-state + operator fallback	TBC
Data integrity	Freshness, completeness, reconciliation and traceability	TBC
Scalability	Expansion/new areas through configuration	Required
Security	OT/IT segregation, RBAC, least privilege and audit	Required
Usability	Rationale, override, what-if and evidence status	Required
Maintainability	Versioned configuration, deterministic tests and simulation	Required

13. Stockyard Layout, Assets & Evidence Gaps

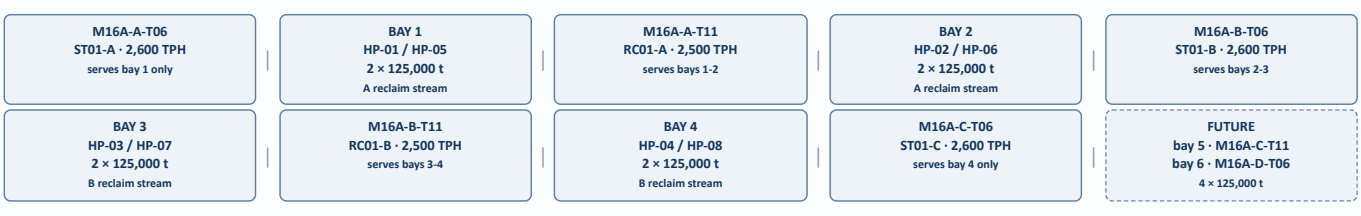
This is the canonical layout/model-coverage summary. Detailed provenance and conflicts remain in §16; positions stay schematic until GA/plot-plan data are received.

13.1 Canonical site / asset register

Site / stage	Material / storage	Stack assets	Reclaim assets	Route	Model status
Louta 102MG mine storage	MINE STOCKPILE-01/-02; **ASSUMED display baseline:** 2 × 100,000 t capacity, 50,000 t t0 each (50%), two-profile 50/50 bands; actual capacity/live state TBC — the 494,274 t ore capacity is *not* theirs	102MG-ST-01/-02 visible	route/rate/reach TBC	intended transfer toward 110LG; never direct to C1	**Reference only:** `MODELLING_GAP[MISSING_ROUTE_RATE_REACH]`; excluded from solver inventory, jobs and KPIs
Louta 110/120LG homogenized ore	2 stockpiles · 494,274 t total · 2 × 247,137 t **DERIVED** equal split · 50% start = 123,568.5 t each (247,137 t total, ASSUMED)	1 shared 110LG-ST-01 · 2,100 TPH wet · chevron via boom conveyor 110LG-BC-01	2 bridge reclaimers 120LG-RC-01/-02 · 3,000 TPH wet each	RB1–RB3 → common RB4/RB5 → washing L1 or L2	**Solver-backed S0 route:** both ore lanes and all three machines are planned; displayed multi-profile bands remain non-spatial metadata
Louta washing / beneficiation	Scrubber residence **2.5–3 min CONFIRMED**; overall washing-line residence **0.5 h (30 min) ASSUMED**	2 parallel production lines **L1 + L2**, 3 Mt tsm/year each; solver process-transfer stages, not mobile machines		recovered WIP → 130LG product stacking	**Solver-backed process transfer:** documented case recovery is applied; no same-bucket passthrough and cumulative matured-mass availability enforce the assumed 30-min dependency
Louta 130/140LG product	383,024 t total · 2 × 191,512 t **planning/solver lines** (DERIVED equal split); starting mass 95,756 t each (50%) — physical pile count not stated in the source	1 shared 130LG-ST-01 (fed by 130LG-CV-11/SL11); **1,000 TPH S0 envelope ASSUMED**	140LG-RC-01/-02 · 3,000 TPH · conservative one-per-planning-line pairing	RL1/RL2 → flip gates 140LG-FG-01/-02 → RL3/RL4 → C1	**Solver-backed S0 route:** stack, product inventory, reclaim and exact 8,000 t C1 contribution are planned
Benguerir mine-ore storage	`BG-MINE-1/2`, 2 × 200,000 t **ASSUMED S0 capacity**, 100,000 t start each; deliberately larger than Louta's 2 × 100,000 t 102MG display baseline	`BG-MINE-ST-1/-2`, one per pile · 2,100/2,500 TPH borrowed envelope	shared `BG-MINE-BWR` · 2,100/2,500 TPH borrowed envelope	`TM1` / `TSM2` / `TSM3` handoffs → mine piles → screened transfer	**Solver-backed:** inventory and stack/reclaim jobs planned; names, reaches and rates are assumptions pending tags/GA
Benguerir screened/raw storage	`BG-RAW-1/2`, 2 × 200,000 t (400,000 t Table-20 total), 100,000 t start each · <15 mm	`BG-RAW-ST-1/-2`, one per pile · 2,100/2,500 TPH	shared `BG-RAW-BWR` · 2,100/2,500 TPH	three washing campaigns; raw-train interface TBC	**Solver-backed:** both piles and all three machines participate in conserved process transfers
Benguerir washing / beneficiation	Scrubber residence **3 min CONFIRMED**; overall line residence **0.5 h (30 min) ASSUMED and modelled**	`BEN-WASH-L1/L2/L3`, all three OPERATING/configurable in S0; process stages, not mobile machines		recovered WIP → shared ST3 → B02	**Solver-backed:** three demand-driven feed/recovery campaigns; rates, recoveries and grade mapping are borrowed/ASSUMED
Benguerir B02 washed	`BG-P1/P2`, 2 × 191,512 t · settled product	current twin-boom `ST3` · 2,100/2,520 TPH	current `RC2/RC3` · 2,600/3,120 TPH; S0 pairs RC2→P1 and RC3→P2	C2 → M16; product train	**Solver-backed:** ST3 stack, pile inventory and RC2/RC3 reclaim jobs are scheduled from demand
M16 humid	8 × 125,000 t in **4 side-by-side two-hangar bays** (+ 4 × 125,000 t future); hangar names/left-right order INFERRED	ST01-A/B/C · ≈2,600 · one per line `M16A-A/B/C-T06`	RC01-A/B · ≈2,500 · `M16A-A/B-T11`	2 separate gate banks `M16A-A/B-H01…H04` → PAP or drying (confluence TBC)	Modelled; each shared line visibly spans its horizontal pile pair; machine reach remains bay-adjacent
M16 dry	2 × 50,000 t	trippers	gravity gates	rail→SPH	Interface abstraction
M17 open/closed	2×100,000 / 1×100,000 t	ST01 / T04T	manual / RC01 ≈1,100	melting	Manual open; closed modelled

Layout rulings: M16 is **four two-hangar bays with the machine lines interleaved between them** — `M16A-A-T06` (ST01-A) · bay 1 · `M16A-A-T11` (RC01-A) · bay 2 · `M16A-B-T06` (ST01-B) · bay 3 · `M16A-B-T11` (RC01-B) · bay 4 · `M16A-C-T06` (ST01-C) — per `Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3. In the mimic, each bay is explicitly drawn as **one side-by-side horizontal pair**, with every serving machine line spanning both hangars (project-team display ruling, 26-Jul-2026); this makes shared access legible without changing model reachability. There is **no shared T06/T11 corridor** and no detached machine corridor. A machine translating between adjacent bays serves **only** those bays (project-team rule, 25-Jul-2026): ST01-A→bay 1 · RC01-A→bays 1-2 · ST01-B→bays 2-3 · RC01-B→bays 3-4 · ST01-C→bay 4; an unreachable job is rejected by validation and by the ASP work-area rule. Same-pile stack/reclaim is permitted subject to separation and work-zone rules, with numerical values TBC. Hangar names and left/right order remain INFERRED; metric coordinates remain schematic (the PFD is a flow sheet, not to scale); mine-site coordinates and concurrency applicability remain TBC.

13.2 M16 — side-by-side hangar pairs with machine lines between bays



Reclaim take-off — two separate streams: each reclaimer feeds its own `T11` → `T11S` magnetic separator → `T12` (belt scale) → hopper `M16A-A-R01` with air blasters → **its own clamshell + rod gate bank** (`M16A-A-H01…H04` / `M16A-B-H01…H04`) → `T13` outbound belt at 1,500 TPH. **That chain is confirmed equipment.** What is *not* confirmed is the destination: neither the PFD nor PRO-0001 says which gate or downstream diverter sends a stream to PAP rather than to drying, and the confluence point is not resolvable from the available sheet. The model therefore separates the two — a **confirmed** gate-bank link per stream, then a **derived/TBC** route selection toward PAP (six parallel 1,500 TPH conveyors, 3 running + 3 future) or drying (two parallel 1,299 TPH conveyors, both out of scope). One destination per bank per bucket is derived from the single outbound belt; the documented 10-minute changeover is recorded but **not modelled**.

Reception envelope — one current path: PRO-0001 §2.1.1 places auto sampler `M16VP01` on `M16A-A-T01` and `M16VP02` on "the adjacent **future** conveyor", and the PFD marks that line FUTURE. C1 and C2 therefore **share one current 2,600 TPH reception path**: aggregate reception is capped at 2,600 TPH, never 2 × 2,600 TPH, and the initial in-transit profile shows the corridors running alternately rather than both at full rate in the same hour.

> **What is confirmed and what is not:** the eight 125,000 t hangars, the four-bay/machine-line sequence, the machine identities, their lines and their rates are **confirmed** by `Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3. The **hangar names inside a bay are inferred** — the PFD does not name them, so `HP-01…08` remains a working convention pending the plot plan and the official SMS pile-ID register. Metric positions stay schematic: the PFD is a flow sheet, not to scale. Machine-to-bay assignment applies the project-team rule that a machine translating between adjacent bays serves those bays only; per-line travel limits still require a surveyed GA.

13.3 Benguerir — six physical piles with three solver-backed washing campaigns



The six physical stockpiles are `BG-MINE-1/2` , `BG-RAW-1/2` and `BG-P1/P2` . The three hopper rows and line WIP buffers are finite solver interfaces, **not additional on-ground stockpiles**. The mine-ore, screened and B02 stages are connected at process level but use their own pile and machine resources. B01 `04BG-*` and B02 `RB/SL/RL-*` physical tags are not merged without PFD reconciliation.

Evidence / issue	Status and consequence
Table 20 screened-yard topology	**User-confirmed** 20-Jul-2026; retained as `BG-RAW-ST-1/-2` + `BG-RAW-BWR` and promoted into the planning model
Six-pile capacity / starting mass	**Physical S0 spaces:** `BG-MINE-1/2` = 200,000 t capacity and 100,000 t start each (**ASSUMED**); `BG-RAW-1/2` = 200,000 t capacity and 100,000 t start each (400,000 t Table-20 total); `BG-P1/P2` = 191,512 t each. The first mine-ore pair is intentionally larger than Louta's assumed 100,000 t 102MG piles. Timestamped SMS telemetry is not loaded.
Three mine hoppers	`TM1` , `TSM2` and `TSM3` are represented by three finite 10,000 t planning handoffs. This is a campaign buffer, **not a claimed physical hopper capacity or live reading**; exact tags/capacities/routing remain TBC.
Raw-grade and line mapping	Benguerir and Louta share the mine-neutral grade IDs `RAW-BT` , `RAW-BT-HIGH-CD` and `RAW-BT-LOW-CD` ; the mine is not encoded in the grade name. S0 uses L1/L3 BT→TBT and L2 Low-Cd→BT. Chemistry, recoveries and mapping are **ASSUMED simulation inputs borrowed from Louta design cases**, not live Benguerir assays or approved recipes.
Mine-ore machine identities and reach	`BG-MINE-ST-1/-2` and shared `BG-MINE-BWR` are user-directed S0 planning assets. Their 2,100/2,500 t/h envelopes are borrowed; physical tags, surveyed reaches and metric GA coordinates are **TBC**.
Table 20 doc no./rev, remaining tags and coordinates	**TBC**; none invented
2,100/2,500 vs 2,600/3,120 TPH	**Conflicting**; raw→C2 at 2,600 assumes resolution
ST3/RC2/RC3 and **	**Current configuration and solver-backed**; notation only. S0 plans recovered-product stacking through ST3, then pairs RC2→BG-P1 and RC3→BG-P2 for demand withdrawals.
Washing-line planning baseline	`BEN-WASH-L1/L2/L3` are all **OPERATING and configurable** in S0. Commissioning/as-built lifecycle evidence remains TBC.
Washing-step timing	**Scrubber residence = 3 min CONFIRMED** by `Q80130-BEN-B02-PR-BOD-0001` Rev 0, §7.1 p. 19 and §7.3.1 p. 25. The separately instructed **ASSUMED overall line residence = 0.5 h (30 min)** is enforced as the S0 wash-to-ST3 dependency; the 3 min component is not added again.
"3 qualities on 2 piles"	Zoning/campaign/compatibility rule **TBC**
Train routes/station	Interfaces documented; ownership/sharing **TBC**
B01 rates	541+89=630 vs 628; 1,208 vs 1,055 · unresolved

> **Simulator coverage:** all six physical Benguerir piles, all nine mobile stockyard machines, all three mine-hopper handoffs and all three washing campaigns are connected to the planning model. A campaign plans hopper-to-mine stacking, mine reclaim, screened-pile stacking, washing reclaim/recovery and recovered-product stacking through `ST3` ; configured train or C2 demand then creates `RC2` / `RC3` reclaim work. The former continuous `BG-WASH-IN` shortcut is retired and inactive. Hopper and WIP spaces are solver interfaces rather than physical stockpiles. OPERATING and availability are scenario configuration, not live telemetry.

> **Mimic inspection contract: Initial state** shows scenario t0; **Final state** shows terminal plan inventory after all horizon buckets. Stockpile fill length is proportional to physical occupancy rather than stretching grade bands to the pile outline. Hover exposes grade composition, physical mass, capacity/fill, material state and stage-correct TSM equivalent when a conversion profile exists. Hopper and WIP interfaces do not appear as extra physical stockpiles.

13.4 Material and grade states



Material state and grade are separate model dimensions. A washed grade vector must never be reused for raw/screened phosphate.

Stage	Documented physical profile	Chemical / grade status	Model profile
Benguerir mine/screened	<15 mm · 10–14% moisture · 1.2–1.4 t/m³ · 30–35°	Shared mine-neutral raw IDs; S0 selects BT on lanes 1/3 and Low-Cd on lane 2. No live/pile-specific Benguerir assay or cut-offs	`BEN-RAW-SCREENED` · solver-backed BG-MINE-1/2 and BG-RAW-1/2
Benguerir filtered	23% moisture · 1.2–1.4 t/m³ · 35°	Assays/cut-offs TBC; S0 recovered mass derives from borrowed case recoveries	`BEN-WASHED-FILTERED` · finite line WIP interface
Benguerir settled	15–20% moisture · 1.2–1.4 t/m³ · 35°	TBT/BT/MT names; S0 produces TBT/BT, while site limits and approved pile rules remain TBC	`BEN-SETTLED-PRODUCT` · solver-backed BG-P1/P2
Louta mine/product	Raw: 15% moisture · SG 2.9 · F100 10 mm; combined product: moisture <22% · SG 3	Raw BT profiles feed assumed 102MG/110LG display compositions; LT-P1/P2 use assumed washed TBT/BT/MT distributions; operating mapping/cut-offs TBC	Separate raw reference and product-planning profiles

M16 humid	PGNAA + moisture + SMS/SQMS	Grade/SQP IDs/limits TBC	M16-WASHED-HUMID

Demo grade	BPL	Cd ppm	MgO %	SiO_2 %	Moisture %	Status
TBT	74	22	0.8	2.0	12	washed simulation placeholder
BT	68	30	1.1	2.8	12	washed simulation placeholder
MT	63	35	1.4	3.5	12	washed simulation placeholder

Louta product-yard demonstration distribution

Product pile · starting state	Assumed grade distribution	Mass-weighted composite used by planning
LT-P1 · 95,756 / 191,512 t (50%)	60% TBT + 40% BT	BPL 71.6% · Cd 25.2 ppm · MgO 0.92% · SiO_2 2.32% · moisture 12%
LT-P2 · 95,756 / 191,512 t (50%)	60% BT + 40% MT	BPL 66.0% · Cd 32.0 ppm · MgO 1.22% · SiO_2 3.08% · moisture 12%
Yard aggregate · 191,512 / 383,024 t (50%)	30% TBT + 50% BT + 20% MT	BPL 68.8% · Cd 28.6 ppm · MgO 1.07% · SiO_2 2.70% · moisture 12%

TBT/BT/MT are **ASSUMED washed-product demonstration vectors**; their operating Louta grade mapping and cut-offs remain TBC. Band widths show mass shares only—not surveyed spatial order, stacking chronology or reclaim sequence. For these Louta demonstration piles, planning uses each pile's aggregate composite quality and does not select a component profile or zone independently. Profile-selective reclaim requires surveyed boundaries and the active reclaim face; the implemented M16 pile-construction blend is the separate, explicitly bounded case in §13.7. The shared `RAW-BT-*` grades are a **separate material state**. Louta's display compositions remain assumed metadata. In the Benguerir solver chain they are campaign feed identifiers: BT feeds L1/L3 and Low-Cd feeds L2 before the assumed wash transformations produce TBT/BT. Raw-grade vectors are never applied directly to washed-product inventory.

Shared raw BT grades — current chemistry source: Louta L02 design basis

Profile / specification	BPL %	MgO %	CO_2 %	SiO_2 %	Cd ppm	Moisture %	SG	PSD
BT raw — component of both demonstration piles	56.06	1.49	8.21	13.71	21	15	2.9	F100 10 mm
High-Cd BT raw — P1 component	55.11	1.50	8.41	16.28	25	15	2.9	F100 10 mm
Low-Cd BT raw — P2 component	58.28	1.13	8.75	10.22	13	15	2.9	F100 10 mm
Combined product specification — reference only	68.29	<0.6	6.81	<4	17	<22	3	—

Louta display-only composite chemistry and Benguerir S0 feed assignments

Raw pile · assumed mass composition	BPL %	MgO %	CO_2 %	SiO_2 %	Cd ppm	Moisture
102MG-SP-01 · 50% BT + 50% High-Cd	55.585	1.495	8.31	14.995	23	Louta 15%
102MG-SP-02 · 50% BT + 50% Low-Cd	57.17	1.31	8.48	11.965	17	Louta 15%
110LG-OL1 · 50% BT + 50% High-Cd	55.585	1.495	8.31	14.995	23	Louta 15%
110LG-OL2 · 50% BT + 50% Low-Cd	57.17	1.31	8.48	11.965	17	Louta 15%
BG-MINE-1 / BG-RAW-1 · BT S0 campaign	56.06	1.49	8.21	13.71	21	Benguerir 10–14% physical metadata
BG-MINE-2 / BG-RAW-2 · Low-Cd S0 campaign	58.28	1.13	8.75	10.22	13	Benguerir 10–14% physical metadata

Source: `Q80130-LOU-L02-PR-BOD-0001` Rev B, 12-Apr-2023, §6.1 pp. 14–15 and p. 18, plus project-team multi-profile and shared-grade clarifications through 28-Jul-2026. The grade IDs are shared across the mines, but the numeric chemistry used for the Benguerir S0 campaign is borrowed from the Louta design basis and remains **ASSUMED**, not a Benguerir assay. Louta's R-band shares remain schematic display metadata; Benguerir's solver-backed pile state uses its configured campaign grade and live/terminal tonnage. The combined-product row retains the document's own stage wording and does not replace the TBT/BT/MT demonstration vectors.

Needed: pile survey or SQMS segment data to replace the assumed shares, band widths and mapping; live/pile-specific Benguerir raw assays and raw/washed min-typ-max assays by site/grade; Grade/SQP IDs and cut-offs, sampling basis, PSD/moisture, pile/zone campaign/compatibility rules and SMS mappings. Live SMS/SQMS values override the demonstration composition and current design-basis vectors.

13.5 Louta — four stages from mine storage to the C1 corridor



Applied planning geometry, not documented pairing: the process description confirms two ore stockpiles and two ore reclaimers but never states which reclaimer serves which pile — both reclaim routes converge on the common RB4/RB5 transfer that can feed either washing line. The mimic therefore draws one reclaimer per pile as the applied planning arrangement. The same caution applies downstream: the **physical** product-pile count is not stated, so LT-P1/LT-P2 are planning/solver lines with a derived equal split.

> **Mimic inspection contract (Louta): Initial state** shows 102MG mine storage at 50,000 t per pile, 110LG ore at 123,568.5 t per pile and product at 95,756 t per line — all 50%. **Final state** shows terminal plan inventory for the solver-backed 110LG and product lines. Only 102MG stays at its static assumed reference baseline in both views.

Authoritative equipment source: **Louta Beneficiation Plant Process description, Rev B (IFA), 25 April 2023, §3.1 pp. 5–8. Document-control inconsistency recorded:** the cover states `Q80130-LOU-L02-PR-REP-00002` while every body header states `Q80130-LOU-L02-PR-REP-00004`; both are retained and neither is silently preferred. Storage totals come from the **Louta BoD Rev B**, which gives no individual pile capacities. BoD p.20 lists one ore and one product reclaimer, while the later tagged process description identifies `120LG-RC-01/-02` and `140LG-RC-01/-02`; S0 uses the tagged two-plus-two configuration and preserves both count conflicts. Neither PDF contains `102MG` or the label `C1` (the process description calls the outbound route `BL`).

> **S0 stocked-route policy — ASSUMED project policy, not a design-document operating rule.** Exact configured targets are C1 dispatch **8,000 t**; recovered output **4,000 t from L1 + 4,000 t from L2**; terminal `110LG-OL1/-OL2` **123,568.5 t each**; terminal `LT-P1/-P2` **95,756 t each**; and line WIP **0 t**. L1 uses documented BT feed/recovery **584 t/h / 68.22%**; L2 uses Low-Cd **588 t/h / 67.71%**; High-Cd **579 t/h / 68.83%** remains documented but unselected. BT→L1→TBT and Low-Cd→L2→BT are **ASSUMED mappings**. `MACHINE_COVERAGE_REWARD=0` and the positive commanded-tonne cost remove coverage rewards and token jobs.

The S0 source interface uses two finite **10,000 t ASSUMED mine-handoff inventories**; they are not 102MG. Common RB4/RB5 is conservatively capped at 588 t/h with one ore reclaimer at a time; the unresolved RL3/RL4/C1 confluence permits one product reclaimer at a time. The 130LG stacker envelope is **1,000 t/h ASSUMED** because no source rate is supplied. Direct ore and product storage bypasses are documented, so idle equipment may be legitimate. Solver output reports scheduled work or structured dispositions: unavailable, no inventory, no target/free capacity, missing route, shared-route peer selected, or target met without that machine. Neither reclaimer pair is identified as duty/standby in the PDFs.

Stage	Known	Not established	Required next
102MG mine storage	MINE STOCKPILE-01/-02 and visible 102MG tags; pre-beneficiation and never direct to C1. S3-A25 displays an assumed 100,000 t capacity / 50,000 t t0 per pile	Actual capacity/live inventory, grade order, duties, reach, route/rate and metric positions. No supplied Louta PDF contains 102MG or establishes its connection to 110LG	Full drawing, GA, route/rate/reach matrix and timestamped SMS/SQMS snapshot; remains `MODELLING_GAP[MISSING_ROUTE_RATE_REACH]`
110/120LG homogenized ore storage	Two solver-backed lanes; shared 110LG-ST-01 at 2,100 TPH wet; 120LG-RC-01/-02 at 3,000 TPH wet; common RB4/RB5; total 494,274 t	Equal pile split DERIVED; start/terminal state, grade bands and conservative pairing ASSUMED. BoD says one ore reclaimer; later tagged description says two, which S0 uses	Live stocks/assays, pile schedule, reach/GA, common-route capacity/concurrency and approved campaigns
Washing L1 / L2	Two 3 Mt TSM/year lines. BT 584 t/h / 68.22%; High-Cd 579 / 68.83%; Low-Cd 588 / 67.71% are documented. Scrubber 2.5–3 min CONFIRMED; overall 30 min ASSUMED. S0 models recovery-bearing transfers with no same-bucket passthrough, cumulative matured-mass availability and a minute dependency	BT→L1→TBT and Low-Cd→L2→BT are ASSUMED; availability, residence validation and shared-tank/filter constraints remain TBC	Operating timestamps, output plan, availability and approved campaign/concurrency rules
130/140LG product storage	383,024 t; solver-backed 130LG-ST-01 and 140LG-RC-01/-02; documented RL1/RL2 → flip gate → RL3/RL4 → outbound BL	Physical product arrangement, 130LG/SL11 rate (1,000 TPH S0 envelope ASSUMED), pairing and confluence TBC. BoD says one product reclaimer; tagged description says two. C1 is a project/model binding, not a PDF label	Arrangement, live SMS/SQMS, stacker rating, reach matrix, confluence capacity and operating grade map
Option boundary	Grey `TO DIGUE (OPTION-2)` exists	Phase-1 selection	Formal option decision; excluded meanwhile

13.6 Control domains

Domain	Coordinated scope	Why separate
A · Phosphate	Louta+C1+Benguerir+C2+M16→PAP/drying/SPH; one plan, three SMS outputs	Shared quality, inventory, corridors and M16 resources
B · Sulphur	M17 closed-store replenishment/reclaim→melting	No shared material, machines or blend constraints

Both use one configurable engine and UI but solve independently. New areas/SMS are added through configuration.

13.7 Current implementation baseline — September 2026

This subsection records the current application behaviour and supersedes any earlier prototype wording in this scope that implies a single-day horizon, fixed one-hour job duration, a separate contingency-entry table or direct two-stream reclaim blending.

Area	Implemented behaviour
Planning envelope	Default tactical horizon is **7 days / 168 hourly decisions**. Optional rolling refinement uses 15-minute decisions in the first four hours and hourly decisions thereafter. Outage, derate and execution boundaries retain exact minutes, and adjacent compatible slices are merged into one continuous job rather than displayed as artificial one-hour jobs.
Demand	One dated seven-day table covers Benguerir, Louta and MPH. A unit may have **multiple grade/quantity campaigns on the same day**; each row carries an intraday start/end and priority. Demand is retained when a route is unavailable and is reported as unmet rather than silently removed.
Routing and contingencies	Plan → Routing availability reuses the Monitor's global flow diagram with All/Louta/Benguerir/MPH filters. Clicking a planner-controlled machine, conveyor, route or plant opens one harmonised control: **Available / Disabled / Future**, plus a physical-rate cap, applied for the whole run or an exact timed interval. This consolidates planned contingencies; a separate table-entry page is no longer required. Current state is explicit on the icon/line and in the consolidated control list.
Configuration	The simplified Configure section includes one Grade configuration page with Chemical grades, Yield & moisture, Material profiles and Receiving limits. Grade records can be added or deleted persistently; a referenced grade must first be remapped, preventing orphaned material profiles or stockpile assignments. The user-facing nominal grade name is **BT**. PAP-2 exposes the same three lifecycle states as every destination and defaults to **Future**.
Mass and quality basis	Capacities, inventories, rates, jobs and blend tonnes are physical/as-handled values at the applicable stage. Raw physical mass → TSM uses `(1 − raw moisture) × beneficiation yield`; wet washed product retains configured moisture; after drying, 1 physical dry tonne = 1 TSM. Chemistry of a blend is dry-solids weighted, while moisture is physical-mass weighted. TSM appears as a secondary label/hover value only when a valid grade conversion exists.
Integrated mimic and inspection	The same diagram is used for planning and monitoring. Reduced labels preserve legibility while machines, plants and conveyors carry prominent state cues. Stockpile fill is drawn proportional to physical occupancy; hover exposes stockpile grade composition, physical mass, capacity/fill, stage and applicable TSM equivalent.
Job schedule and results	Monitor exposes readiness, demand/service and unmet quantity, quality/off-spec/give-away, inventory/safety/capacity, route and machine utilisation/coverage, positioning/conflict, and replanning/recovery evidence. It also shows the active configurable objective weights. Schedule filters include site/domain, machine, operation and status plus **Displayed period = full run or an exact calendar day**.
M16 blending	The implemented point is the **M16 humid stockpile HP-07** before its PAP receiving campaign. The dedicated Blending page offers **Blend matrix** (operator-set fixed shares) and **Optimized shares** (component masses are solver variables). In optimized mode composition, stacking time, routes, equipment calendars, source inventory, competing work and destination interval are solved together in physical tonnes. The proposal may change on replan until stacking starts; executed stacking locks the resulting matrix. Reclaim cannot start before the configured pile-ready boundary, and stacked batch mass must equal matrix mass exactly.

The following blending extensions are **not yet implemented**: Benguerir washed-product piles into a train campaign; multi-quality raw-stock feed into washing lines; and spatial tracking of layers/zones and their changing reclaim composition. Grade-class chemistry is currently configured as vectors plus destination receiving intervals; defining the grades themselves entirely by min/max characteristic intervals remains a planned modelling extension.

> **Verification status:** the focused blending/scheduling regression set passes 61 tests. A full seven-day physical solve has also returned a feasible 12,000 t HP-07 campaign with jointly selected 1,000 t TBT + 11,000 t BT-LOW-CD and no reclaim before the 48-hour ready boundary. These are deterministic simulator checks, not proof of plant performance. The wider legacy repository suite still contains stale UI/report expectations and is not yet a clean acceptance suite; operational validation continues to require governed SMS/SQMS data, approved quality intervals, route evidence and shadow/pilot comparison.

PART II — FUNCTIONAL DESIGN (Domain Model · Allocation Decision Model · Engineering Basis)

14. Domain / Data Model (planning digital twin)

The layer keeps a **planning abstraction**, continuously refreshed from the authoritative SMS/SQMS 3D inventory and machine state, then augmented with demand, routes, recipes and jobs. It never replaces the physical twin. The canonical three-site value chain, system boundary and assets are defined once in scope §§1, 4 and 13; this part defines only the information and optimisation model.

14.1 Entities

Entity	Key attributes	Source
Site	SMS endpoint, role, spaces and assets	configuration + SMS
StockSpace / Inventory	capacity/state; material profile; aggregate planning quality; optional grade-distribution metadata; 1-m segment mass, quality and age	SMS/SQMS + planning
MaterialProfile	process state (`raw_screened` , `washed` , `dry`), physical envelope and evidence	engineering + master data
QualityClass	stage-compatible chemistry, receiving intervals and anti-contamination matrix	operations master data
Machine	rate, reachability, rail, travel/setup, separation, calendar, state and position	SMS state + configured capability
RouteGraph	nodes/links/gates, rates, conflicts, transitions, delays and valid paths	PFD/GA + planning
Destination / Demand	dated grade campaign, physical tonnes/rate, intraday window and priority; several non-overlapping campaigns may target one unit in one day	ERP/operations
Job	machine, space, type, route, pattern, rate, tonnes, timing and status	planning output
BlendCampaign	target stockpile, eligible incoming grades, fixed matrix or optimised shares, physical batch tonnes, readiness and execution-lock state (§15.6)	operator configuration + planning output
GradeCostModel	grade-dependent consumption and rock value; data not yet received	process engineering/finance

14.2 Route graph (layer-owned) — versus the ASP "Route"

ASP “Route”	Route graph owned by this layer
`(Device, JobType, free-text Location)` ; filters plan-editor choices	Physical nodes/links/gates; capacities/minimum rates; delays/in-transit mass; states, conflicts, transitions, availability and valid source→destination paths

Every emitted job passes the physical graph first; ASP is a **second gate**. PFD/TBE data define connectivity and rates; GA/plot-plan data are still required for metric travel and transition times.

14.3 Entity relationships



14.4 Phase 1 instances (from the scope note)

Domain / site	Planning role	Solver coverage	Canonical evidence
Louta	110LG ore stock → L1/L2 recovery → 130/140LG product yard → C1	**Stocked route solver-backed from 110LG through C1; only 102MG remains reference-only**	Scope §13.1/13.5; evidence §16
Benguerir	Mine storage → raw storage → three washing campaigns → washed-product yard → C2/train	Six stockpiles and nine mobile machines are solver-backed; washing lines are recovery-bearing process stages	Scope §13.3; evidence §16
MPH M16	Receive, construct humid-product blends, buffer and reclaim → PAP/drying	Current bays, routes and machines modelled; PAP-2 is configurable and Future by default	Scope §13.1; evidence §16

Domain / site	Planning role	Solver coverage	Canonical evidence
MPH M17	Open/manual → closed → melting	Domain-B abstraction; open reclaim exogenous	Scope §13.1; evidence §16

Exact capacities, assets, statuses and source conflicts are intentionally not repeated here; scope §13 is the asset summary and §16 the authoritative evidence register.

15. Allocation Decision Model

Each planning cycle selects mass flows, assets, routes and timing to meet on-grade demand while respecting the coupled three-site constraints.

15.1 Decision variables

Variable	Meaning	Domain
`m[mine, c, t]`	mine reclaim rate onto its **dedicated corridor** `c` in bucket *t* (TPH), with quality	≥ 0
`x[s, d, t]`	reclaim **mass flow** from space *s* to destination *d* in bucket *t* (TPH) — **the primary blend variable**	≥ 0
`delivered[d, t]`	total feed to destination *d*: `delivered[d,t] = Σ_s x[s,d,t]` (TPH)	derived
`unmet[d, t]`	**unmet-demand slack** for destination *d* (TPH) — see eq. (4)	≥ 0
`f[s, d, t]`	blend fraction — **derived, reporting only**: `f[s,d,t] = x[s,d,t] / delivered[d,t]`	[0,1], Σ = 1
`a[reclaimer, s, t]`	reclaimer assigned to a space	{0,1}
`y[stacker, s, pattern, t]`	stacker building a space with a pattern	{0,1}
`b[p, g]`	whole physical tonnes of incoming grade *g* assigned to blend campaign/pile *p*	integer ≥ 0
`ready[p]`	exact minute at which construction of campaign *p* is complete and reclaim may begin	integer minutes
`g[gate/route, t]`	gate / route-path selection (per reclaim stream: which clamshell gate bank routes toward PAP or drying)	{0,1}
`seq / start[job]`	job ordering & timing	—

Flow and fraction are not independent. The model is solved in **mass flows** `x` (linear); blend fractions satisfy the identity `x[s,d,t] = f[s,d,t] · delivered[d,t]` and are recovered after solving. Treating `f` and the delivered rate as independent variables would decouple recipe from throughput and make the quality constraints nonlinear.

15.2 Objective — minimise a weighted sum

Term (weight)	Drives	Priority
w_1 unmet demand `Σ unmet[d,t]·Δt` (eq. 4)	**Feed continuity — no starving PAP / molten-S; w_1 dominates all other weights**	**highest**
w_2 off-spec deviation	**On-spec grade**	**highest**
w_3 downstream process cost *(grade-dependent)*	Placeholder for governed PAP consumption curves; **inactive (0) in the implemented baseline**	inactive
w_4 rock value consumed (give-away)	Don't waste scarce high-grade rock beyond its economic return	high
w_5 machine moves / changeovers / start-stops	Efficiency, wear	medium
w_6 safety-stock violation	Strategic stock cover	medium
w_7 handled-flow / utilisation reward *(maximise)*	Optional throughput/coverage preference; **inactive (0) by default**	configurable

The active scenario exposes the implemented weights beside the Results KPIs: w_1 unmet demand, w_2 off-spec deviation, w_4 quality give-away/rock value, w_5 moves/changeovers, w_6 safety-stock violation and w_7 utilisation. They are operator-configurable and persisted with the planning assumptions. w_3 remains inactive until validated GradeCostModel curves are supplied; the interface does not imply an economic optimum that the available data cannot support.

S0 Louta objective ruling. Machine coverage is disabled (`MACHINE_COVERAGE_REWARD=0`), and a small positive cost is applied to every commanded tonne (`FLOW_TONNE_PENALTY=1`). Exact stocked-route constraints create the work; the objective removes unnecessary excess and tie-driven token jobs. This is an **ASSUMED demonstration policy**, not a rule found in either Louta design document.

15.3 Dimensional formulation — units, mass balance, quality, delay

Conventions. The operator baseline is a civil week: **T = 168 buckets of Δt = 1 h**. An optional rolling-refinement mode uses 15-minute decisions in the first four hours and an hourly tail. Outage, derate, readiness and actual-event boundaries retain their exact minutes; therefore emitted jobs are continuous variable-duration runs and are not forced to one-hour length. Every stockpile capacity, inventory, machine/conveyor/plant rate, demand and scheduled job is expressed in **physical/as-handled tonnes or t/h** at that material stage.

(1) Space mass balance — for every space *s* and bucket *t* (dimensionally consistent: tonnes = tonnes + h·t/h):

```
inventory[s, t+1] = inventory[s, t]
                  + Δt · stacked_rate[s, t]
                  − Δt · reclaimed_rate[s, t]              [tonnes]

0 ≤ inventory[s, t] ≤ capacity[s]
reclaimed_rate[s, t] = Σ_d x[s, d, t]
```

Initial condition: `inventory[s, 1]` = live SMS snapshot (`SpaceValueResponse`). *Horizon boundary:* terminal stock `inventory[s, T+1]` is held ≥ the safety-stock target (soft, w_6) so the optimiser cannot "win" by emptying the yard at the end of the horizon.

(2) Delivered flow & blend coupling — solve in mass flow, report in fractions:

```
delivered[d, t] = Σ_s x[s, d, t]                          [TPH]
x[s, d, t] = f[s, d, t] · delivered[d, t]   (identity; f recovered post-solve)
```

(3) Mass-weighted quality — per destination *d*, parameter *k*, bucket *t* (linear in `x` ; enforced **per bucket**, i.e. continuous compliance, not a horizon average):

```
Σ_s q_k[s, t] · x[s, d, t]  ≥  q_k_min[d] · delivered[d, t]     (min specs, e.g. BPL)
Σ_s q_k[s, t] · x[s, d, t]  ≤  q_k_max[d] · delivered[d, t]     (max specs, e.g. Cd)
```

`q_k[s, t]` comes from the planning twin (1-m segments aggregated over the planned reclaim cut). When stacking modifies a space, its quality updates mass-weighted: `q_k[s,t+1] = (inventory·q_k + Δt·stacked_rate·q_k_in) / (inventory + Δt·stacked_rate)` . For an M16 constructed blend, chemistry is weighted by **dry-solids mass** and moisture by physical mass, so a wetter tonne is not treated as containing the same solids as a drier tonne.

(3a) Stage-aware physical ↔ TSM conversion — one editable yield/moisture profile is selected per merchant grade; no size-cut variant is inferred:

```
raw_to_TSM       = raw_physical × (1 − raw_moisture) × beneficiation_yield
wet_product      = TSM / (1 − wet_product_moisture)
wet_product_TSM  = wet_product_physical × (1 − wet_product_moisture)
after drying:      1 physical dry tonne = 1 TSM
```

Yield loss occurs through washing/beneficiation; retained water explains why wet-product physical mass remains above TSM. The planning balance never substitutes TSM for the physical capacity or rate of handling equipment.

(4) Demand with unmet slack — no hard `delivered ≥ demand` constraint (a hard constraint plus a penalty on the same quantity is redundant and makes tight periods infeasible):

```
delivered[d, t] + unmet[d, t] ≥ demand[d, t],    unmet[d, t] ≥ 0
objective += w₁ · Σ_{d,t} unmet[d, t] · Δt       (w₁ ≫ every other weight)
```

The slack guarantees the model stays feasible when demand physically cannot be met; the **dominant penalty** ensures any feasible way of serving demand is always preferred, and the reported `unmet` is an honest feed-gap signal.

(5) Corridor transport delay & in-transit inventory — with `L_c = round(delay[c]/Δt)` whole buckets (Δt = 1 h → provisional L_C1 ≈ 2, L_C2 ≈ 4; Louta/C1 nearer and Benguerir/C2 farther is user-confirmed; exact metric values are TBC):

```
MPH_arrival[c, t] = mine_dispatch[c, t − L_c]              for t > L_c
MPH_arrival[c, t] = belt_profile₀[c, t]                    for t ≤ L_c
in_transit[c, t+1] = in_transit[c, t] + Δt · (mine_dispatch[c, t] − MPH_arrival[c, t])
```

`belt_profile₀` is the **material already on the belt at the start of the horizon** (mass *and* quality along the belt, from mine-SMS dispatch history / belt scales) — it arrives during the first `L_c` buckets regardless of any new decision. **Dispatch time and MPH arrival time are distinct**: a mine job dispatched at *t* only becomes MPH stock at *t + L_c*; the quality arriving at MPH at *t* is the quality dispatched at *t − L_c*. Mine jobs must **lead** MPH needs by `L_c` .

(6) Louta process transfer, recovery and residence — L1/L2 are transformation stages, not mobile machines:

```
recovered_output[line] = recovery[case] × ore_feed[line]
WIP[line, t+1] = WIP[line, t] + recovered_output[line, t] − product_stacked[line, t]
product_stack_start ≥ ore_reclaim_finish + 30 min
```

The BT, High-Cd and Low-Cd feed/recovery pairs (**584 t/h / 68.22%**, **579 t/h / 68.83%**, **588 t/h / 67.71%**) are documented design cases. The **30-min overall residence is ASSUMED**; only the scrubber's 2.5–3 min is confirmed by the BoD and is included within, not added to, that total. Same-bucket passthrough is prohibited. A cumulative matured-mass constraint permits product stacking only after enough washing output has completed its execution window plus the 30-min delay; transient WIP may carry between hourly boundaries and must finish the horizon at 0 t.

15.4 Constraints

Constraint	Rule / formalisation	MPH source
Corridor (dedicated)	Louta only on `C1` , Benguerir only on `C2` ; flow ≤ 2600 TPH each	dedicated per mine
Louta S0 purpose targets	When enabled: exact C1 dispatch = **8,000 t**; exact recovered output = **4,000 t per L1/L2**; exact terminal `110LG-OL1/-OL2` = **123,568.5 t each**, `LT-P1/-P2` = **95,756 t each**, and line WIP = **0 t**. L1 BT→TBT and L2 Low-Cd→BT are **ASSUMED** mappings.	project-team demonstration policy; not in design PDFs
Louta shared paths	Common RB4/RB5 limited conservatively to **588 t/h** and one ore reclaimer per bucket; unresolved RL3/RL4/C1 confluence permits one product reclaimer per bucket. `130LG-ST-01` uses a **1,000 t/h ASSUMED S0 envelope** because no stacker/SL11 rate is supplied.	process-description topology + conservative assumptions
Corridor transport lag	eq. (5): arrival(t) = dispatch(t − L_c); **C1/Louta NEARER, provisional Δ_C1 = 2 h 05; C2/Benguerir FARTHER, provisional Δ_C2 = 4 h 10** (30/60 km @ 4 m/s working values); initial full-rate `belt_profile₀` ≈5,400 / ≈10,800 t. Relative order user-confirmed 22-Jul-2026; exact lengths/speeds/dwell TBC.	project team + provisional 30 / 60 km · 4 m/s
Mine stock	mine reclaim ≤ mine-yard inventory; mine build follows its plan. Benguerir contention is **stage-internal**: raw outlets share upstream reclaim capacity; washed outlets share RC2/RC3. Raw and washed stages share no piles/reclaimers (see §§13.2, 16).	mine SMS / B02 BoD
End-to-end continuity	mine ship + in-transit + MPH buffer sustain the MPH→PAP feed over the horizon	chain coupling
Mass balance & capacity	eq. (1) per space/bucket, with initial + terminal conditions	physical
Blend / grade spec	eq. (3): mass-weighted quality per bucket (`Σ q·x ≥ q_min·delivered` , `Σ q·x ≤ q_max·delivered`)	quality specs; Cd for SPH
Machine rate, reach & stability	`min_stable_rate·run ≤ x ≤ max_rate·run` (semi-continuous); only **reachable** (machine, space) pairs — machine-to-pile reachability from configured **work areas**	machine data / work areas
Machine positioning & setup	The post-solve minute schedule inserts a separate **non-material** cross-space activity before the next job: estimated translation time (job-range midpoint→midpoint at configured travel speed) + setup time. A **green** bar is an exact reserved minute interval; the machine may position early and wait around a blocked calendar. A **red positioning conflict** means no valid interval was reserved and targeted replanning is required; that exact-profile job is withheld from ASP plan-item export until timing validates. Positioning carries no tonnage/grade/path. Blank time is idle/unallocated. First-job positioning and same-space face progression are not separate bars.	current geometry/speed/setup are ASSUMED; obtain plot plan / GA, surveyed common rail frame, live position and operating travel/setup data
Same-device exclusivity	a device executes **one job at a time**; no overlapping jobs (ASP ERROR)	ASP Same-Device Overlap
Reclaimer↔space exclusivity	a reclaimer serves **one space at a time, movable** (`Σ_s a ≤ 1`)	confirmed movable
Concurrent stack+reclaim on a space	**PERMITTED at MPH M16 (project-team confirmation, 15-Jul-2026)**: stacking and reclaiming may occur simultaneously on the same stockpile, subject to spatial separation, work-zone, route, geometry and ACS/PLC constraints; the scheduler prevents collisions and overlapping work zones and must **not** prohibit same-pile concurrency categorically (no fixed A/B double-buffering rule). **Exact numerical safety limits remain TBC** (YardMaster/ASP configuration, ACS specification, machine supplier data); applicability at Louta/Benguerir stays configurable/TBC	project team 15-Jul-2026 · ASP UC-28
Spatial minimum separation	machines on the same (or, per `CurrentPileOnly` switch, any) pile keep ≥ the **configurable minimum distance** (1-D X positions)	ASP Spatial Conflict (WARNING)
Maintenance / work calendars	no job in a global or device-specific calendar window (recurring entries expanded ≤ 1 year)	ASP Work Calendar (ERROR)
Whole-run and timed controls	machines, conveyors, washing lines and receiving plants obey their selected lifecycle/availability state and physical rate; timed outage or derate windows split capacity at exact-minute boundaries	operator scenario + contingencies
Routing / gate (graph)	every job on a **valid source→destination path** of the route graph (§14.2): link capacities, min stable rates, gate states (each clamshell gate bank routes its stream toward PAP or drying — one destination per bank per bucket), **route conflicts / shared conveyors mutually exclusive**, transition times between successive gate states	route graph (layer-owned)
Anti-contamination	incompatible qualities not co-stacked in a space	compatibility matrix
Constructed blend readiness	exact grade tonnes close to the campaign batch; eligible-share and receiving-quality intervals hold; target capacity and all incoming routes/resources remain feasible; no reclaim precedes the exact completion minute; executed matrix is immutable on replan	implemented M16 blend campaign
Sulphur (two-stage)	open reclaim exogenous (payloader→ `CLOSED-1`); layer runs `RC01` →melting; **start/stop limits ≤ 8 start-stops/hr** (lime dosing)	payloader manual; 8/hr
Drying-feed cap	humid→dry reclaim ≤ drying-plant intake capacity	drying plant (OOS)
Demand service	eq. (4): `delivered + unmet ≥ demand` with dominant `w₁` penalty — **no separate hard feed constraint**	PAP / molten-S demand
FIFO / material age *(soft)*	prefer reclaiming older material of the same grade first (stacking-age per 1-m segment); avoid degradation of exposed stock	ASP FIFO (INFO) / age analysis

Defense in depth: *every constraint marked with an ASP rule is also enforced vendor-side — plans submitted to ASP are re-validated (* `AspEvaluatePlanResponse` *), an independent second gate (§15.7).*

15.5 Implemented solve and replanning approach

Layer	Horizon / cadence	Solves	Implemented method
Weekly tactical planner	7 civil days · 168 × 60 min by default	demand campaigns, physical mass, machines, conveyors, routes, washing transfers, inventory and M16 blend construction	deterministic **OR-Tools CP-SAT**, decomposed by civil day with carried inventory and plan hints
Rolling refinement	optional 15-min first 4 h → hourly tail	the same coupled decisions with finer near-term timing	mixed-resolution CP-SAT; exact-minute event boundaries retained
Event/replan loop	on outage, derate, state/rate deviation or job feedback	preserve executed work; regenerate feasible remaining work	replay actuals, lock executed blend composition, update calendars/capacities and re-solve

Machines, conveyors, washing lines and receiving plants can be Available, Disabled/Future where applicable, or subject to a physical rate cap for the whole run. Exact timed `UNAVAILABLE` and `RATE_CAP` windows model planned or sudden contingencies. Demand is never silently deleted: an unreachable campaign remains visible as unmet demand.

15.6 Implemented M16 stockpile-construction blending

The current implemented blend point is **construction of an M16 humid-phosphate stockpile before PAP/drying feed**. Several eligible incoming qualities are stacked into the selected stockpile; only after the campaign is complete may one reclaimer withdraw the homogenised campaign toward PAP-1, PAP-2 or DRY-FEED. The model does not claim a direct simultaneous two-stream reclaim blend.

Mode	Operator supplies	Planner decides
Blend matrix	fixed physical mass percentages for each eligible grade	feasible routes, equipment, timing and readiness
Optimized shares	eligible grades and min/max share bounds	integer physical tonnes and percentages **jointly** with inventory, competing stockpiles, routes, machines, calendars, outages, derates and the weekly schedule

For campaign `p`, the exact construction closure is `Σ_g b[p,g] = batch[p]`; each selected grade respects its share bounds and dry-solids-weighted chemistry must satisfy the receiving interval. Scheduled stacking tonnes for every grade equal `b[p,g]`. Reclaim is prohibited before the exact `ready[p]` minute. Status progresses `NOT_STARTED → BUILDING → COMPLETE → DEPLETED`. A proposed but unexecuted matrix may change on replan; the first positive executed stacking movement locks the physical matrix and replanning preserves it.

A focused 12,000 t verification case produced one feasible incumbent of 1,000 t TBT plus 11,000 t BT-LOW-CD and held reclaim until the configured 48-hour readiness boundary. This is **test evidence, not a fixed product recipe**: optimized shares depend on the scenario, quality intervals, stocks and active resource controls. More detailed within-pile layer/zone composition during reclaim, Benguerir washed-product blending to BG-TRAIN and raw-feed blending before washing remain future extensions.

15.7 ASP validation rules — severities, governance & quality limits

The vendor-side severities matter for how the layer treats the second gate:

ASP severity	Rules	Vendor behaviour	This layer
ERROR *(blocking)*	**Parameter sanity** · **Work area** · **Inventory availability** (mass within StartX–StopX + analysis availability + **most-prevalent grade** = requested) · **Same-device overlap** · **Work calendar**	plan item invalid; critical errors raise a notification	modelled as hard constraints (§15.4) — an AUTO item should never draw an ERROR
WARNING *(non-blocking)*	**Device availability / busy** · **Spatial proximity** (1-D min distance, `CurrentPileOnly` switch)	operator may proceed	treated as **hard constraints internally**; **project governance may promote selected ASP WARNINGs to blocking conditions in the external optimiser / dispatch gate** (configuration decision)
INFO *(advisory)*	**FIFO** (older same-grade surface material first, incl. other piles)	advisory only	soft objective term (age/FIFO, §15.4)

ASP quality limitations (why blending is owned here). For a reclaim item ASP checks inventory **mass**, **analysis availability** and the **most-prevalent grade within StartX–StopX** — it does **not** prove that a simultaneous multi-source blend meets **continuous SQP limits**. **One ASP job corresponds to one material grade**; the separate **YardMaster Blending Simulation** may display blend results but **has no role in ASP** (or its validation). The **SQP-versus-operator-defined Sub-Grade behaviour remains an unresolved ICD/configuration question** (the FRS states SQP is outside ASP, yet Sub-Grades built on SQP combinations should reflect in ASP). Consequence: all blend/quality optimisation and continuous spec compliance are computed and guaranteed by **this layer**; ASP validates per-job grade plausibility only.

15.8 Execution interface & closed loop — mapping to the vendor ASP (Digimine-YardMaster)

Our **Job** maps directly onto the vendor's **AspPlanItem** (per FRS Q80130-MZI-M16-TAK-45000183621-DG):

Our model	AspPlanItem field	Our model	AspPlanItem field
Machine	Device	target pile / repère range	Pile + metre-canonical StartX / StopX
job type	JobType (Stacking / Reclaiming / …)	stacking pattern	**Method** (e.g. cone-shell)
QualityClass	Material + Grade	gate / conveyor routing	Source / Destination (**ASP Route** — 2nd gate only; real routing validated on the route graph §14.2)
quantity + rate	PlannedTonnage + Rate	timing	StartTime (end = start + tonnage/rate)

The UI and operating plan display `R_start-R_stop`; the adapter converts those values to physical `StartX/StopX` using the selected pile's approved R0/R1000 calibration. The optimiser and all distance-based rules continue to use metres.

Dispatch loop: the layer submits plan items (`AspPlanItem` telegram) → ASP validates against its 8 rules and returns `AspEvaluatePlanResponse` (second feasibility gate, §15.7) → jobs are **auto time-triggered** to YardMaster job management with operator **Accept/Reject** (human-in-the-loop) → `JobUpdated` **telegrams return actuals** → the layer re-validates and **re-plans** on deviations.

Closed-loop feedback content (`JobUpdated`) and what it drives:

Fed back	Used by the layer for
actual status (job state; Run / Finished / Cancelled …)	job lifecycle: Run sets **actual start**, Finished/Cancelled set **actual end**
actual start / end times	plan-vs-actual deviation tracking
moved mass + material progress	remaining-work estimate; inventory reconciliation with the twin
effective rate (from moved mass)	rate-shortfall detection; recalibrating job-duration models
forecast completion	look-ahead conflict check for successor jobs on the same device/route
current machine position (X/Y/Z)	travel-time estimate for the machine's next assignment
source / target + quality	verification of executed routing and grade against the plan

Deviation → validation → replanning: each `JobUpdated` republishes the linked `AspPlanItem` and re-triggers plan evaluation vendor-side (critical errors raise a notification). Layer-side, configurable thresholds on **rate shortfall, forecast-end slip, quality mismatch, cancellation/fault** first fire the **dispatcher** (re-sequence within the tactical plan); deviations beyond the tactical plan's envelope trigger a **tactical re-solve**. ASP's Inventory Forecasting (60-day projection, depletion alerts) and Planning Deviation Report complement the layer's own KPIs.

15.9 Implemented operator workflow, diagram and results

Area	Current behaviour
Configure → Grade configuration	One coherent page with Chemical grades, Yield & moisture, Material profiles and Receiving limits. The grade catalogue is add/delete editable and persisted in `config/grade_catalog.json`; a referenced grade must first be reassigned. User-facing grade names use the canonical BT naming only.
Plan → Demand planning	One dated weekly table; each plant/unit/day can contain several grade campaigns, each with physical quantity, priority and intraday start/end. Positive windows for the same unit/day may not overlap.
Plan → Blending	Dedicated page, enabled per user-facing humid stockpile. Only implemented points expose a toggle; current implementation is M16 humid-stockpile construction with matrix or optimized-share mode.
Plan → Routing availability	Uses the same global flow diagram as Monitor. Clicking a machine, conveyor, washing line or destination sets whole-run state/rate or an exact timed control. The former Planned contingencies page is consolidated here (a compatibility alias only). PAP-1, PAP-2 and DRY-FEED use the same Available/Disabled/Future selector; PAP-2 defaults Future.
Monitor → global flow	Site filter All/Louta/Benguerir/MPH; reduced labels; explicit machine/plant/conveyor status and rate. Stockpile fill width is proportional to physical occupancy. Hover exposes component grade, tonnes, quality, capacity/fill, material stage and TSM equivalent when a conversion profile exists.
Monitor → Results / Job Schedule	Readiness, solver and objective cards; demand/service, quality, inventory/safety, machine/route coverage and contingency/replan results. The job chart has site/machine/operation/status filters plus Full run or exact-day selection. Touching solver slices are coalesced; real gaps, route changes, positioning and exact-minute events start a new run.

The objective-driver cards show their active scenario weight beside the measure. Current demonstration defaults are:

Driver	Weight	Results interpretation
unmet demand	$w_1 = 1000$	physical tonnes unserved; operational service remains a separate KPI
soft quality deviation	$w_2 = 200$	tonne × quality-unit outside a preferred interval
downstream process cost	$w_3 = 0$	inactive until validated consumption curves exist
BPL give-away / scarce quality use	$w_4 = 5$	tonne × BPL-point above the necessary target
activity changes	$w_5 = 2$	machine moves/changeovers/start-stop burden
safety-stock shortfall	$w_6 = 20$	tonne-buckets below safety target
handled-flow/utilisation reward	$w_7 = 0$	inactive by default; coverage is still reported

Emergency overflow/unstacked-arrival slack uses a protected dominant penalty; `FLOW_TONNE_PENALTY=1`, `MACHINE_COVERAGE_REWARD=0`, and mixed-resolution active buckets use a default penalty of 25. These are model controls, not business KPI targets. Operational KPIs remain delivered-versus-demand/unmet, on-spec delivery, grade give-away, feed interruptions, stock risk/capacity, equipment and route utilisation/coverage, conflicts, solve status/time, replan count/latency/stability, and physical↔TSM traceability.

16. Evidence & Traceability Register

Status vocabulary: **Confirmed** (stated in a referenced document) · **Derived** (computed from confirmed values) · **Inferred** (working hypothesis, confirmation required) · **Conflicting** (two sources / internal inconsistency, preserved) · **TBC** (awaiting document/decision). Confidence H/M/L.

Site / domain	Parameter / rule	Value [unit]	Source · rev · ref	Status	Conf.	Design impact	Owner / action
MPH · M16	humid open storage	8 spaces × 125,000 t (→ 12 future)	PRO-0001 / PFD-0001 [3]	Confirmed	H	yard capacity model	—
MPH · M16	machine lines, bays & reachability	**FOUR two-hangar bays with the machine lines interleaved between them**: `M16A-A-T06` (M16A-A-ST01, 2,600 TPH) · bay 1 · `M16A-A-T11` (M16A-A-RC01, 2,500 TPH) · bay 2 · `M16A-B-T06` (M16A-B-ST01) · bay 3 · `M16A-B-T11` (M16A-B-RC01) · bay 4 · `M16A-C-T06` (M16A-C-ST01). The mimic draws each bay as **one side-by-side horizontal hangar pair**, with every serving line spanning both members (display ruling 26-Jul-2026). **No shared T06/T11 corridor.** Reach follows the project-team bay-adjacency rule: ST01-A→bay 1 · RC01-A→bays 1-2 · ST01-B→bays 2-3 · RC01-B→bays 3-4 · ST01-C→bay 4 — never all-to-all. FUTURE continues bay 5 · `M16A-C-T11` (M16A-C-RC01) · bay 6 · `M16A-D-T06` (M16A-D-ST01) + 4 × 125,000 t hangars	`Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3, IFU, 28-Oct-2025 + project-team reach rule 25-Jul-2026 + display ruling 26-Jul-2026	**Machine identities, rates, lines and the bay sequence CONFIRMED**; hangar names and left/right order inside a bay **INFERRED**; metric positions schematic (flow sheet, not to scale); horizontal pairing is geometry-only and changes no solver input	H	machine set, machine lines, `reachable_spaces`, solver candidates, ASP work-area rule and the mimic	plot plan / surveyed GA + official SMS pile-ID register + per-line travel limits
MPH · M16	same-pile concurrent stack+reclaim	permitted subject to spatial separation, work-zone, route, geometry and ACS/PLC constraints; numerical safety limits TBC; Louta/Benguerir applicability configurable/TBC	project-team confirmation, 15-Jul-2026	Confirmed	H	scheduler concurrency rule (§15.4)	obtain numerical separations (YardMaster/ACS/machine supplier)
MPH · M16	PAP / drying take-off	**TWO separate clamshell + rod gate banks**, one per reclaim stream: `M16A-A-H01…H04` and `M16A-B-H01…H04`. Each stream keeps its own `T11` → `T11S` magnetic separator → `T12` (belt scale `T12W`) → gate bank → `T13` outbound belt at 1,500 TPH. PAP is fed by six parallel 1,500 TPH conveyors (3 running + 3 future); the drying unit by two parallel 1,299 TPH conveyors, both OUT OF SCOPE	`Q80130-MZI-M16-ME-PFD-0001[3]` Rev 3 + `Q80130-MZI-M16-PR-PRO-0001` Rev 2 §2.1	**Gate banks + belt ratings Confirmed** and modelled as the confirmed links `L-H-A` / `L-H-B`; the **destination assignment is DERIVED/TBC** and modelled separately as `L-RTE-{A,B}-{PAP,DRY}` — the earlier '2-of-4: 2→PAP, 2→drying' reading is withdrawn, and confirmed ratings never confirm the assignment. Changeover documented but not modelled	H	route-graph arcs: `HUB:M16-T13-{A,B}` then one PAP and one drying link per bank, mutually exclusive (DERIVED from the single outbound belt)	**PAP/drying route confluence TBC** + gate→conveyor assignment from the downstream sheets
MPH · M16	PAP feed	≈1500 TPH/line; Phase 1: one line	PRO-0001	Confirmed	H	demand model	—
MPH · M16	dry closed storage	2 × 50,000 t → rail → SPH	PRO-0001 / PFD-0001	Confirmed	H	dry-stream buffer	—
MPH · M16	reception analysers	PGNAA (BPL·Cd·MgO·SiO_2·CaO/CO_2) + microwave moisture; autosampler + lab	TEN-0002 [B]	Confirmed	H	quality inputs	—
MPH · M17	sulphur two-stage	open 2 × 100,000 t (payloaders ≈300 TPH, manual) → closed 1 × 100,000 t (RC01 1100 TPH)	PRO-0001 / PFD-0004 [3]	Confirmed	H	Domain B model	—
MPH · M17	start-stop limit	≤ 8 start-stops/h (lime dosing)	PRO-0001	Confirmed	H	dispatch constraint	—
Corridors	C1 Louta→MPH	**NEARER (user-confirmed)** · 2600 TPH · provisional ≈30 km / 4 m/s	capacity: TBE/PFD; proximity: project team 22-Jul-2026	Relative order Confirmed; distance/speed TBC	M	corridor model	obtain TBE/datasheet or surveyed route length + effective speed/dwell
Corridors	C1 transit / in-transit	provisional ≈2 h 05 · ≈5,400 t @ full rate	derived from provisional length/speed	Assumed / Derived	L	lag `L_c` (eq. 5)	replace with measured lag + live belt profile
Corridors	C2 Benguerir→MPH	**FARTHER (user-confirmed)** · 2600/3120 TPH · provisional ≈60 km / ≈4 h 10 / ≈10,800 t	capacity: B02 BoD; proximity: project team 22-Jul-2026	Relative order Confirmed; distance/lag TBC	M	corridor model	obtain TBE/datasheet or surveyed route length + effective speed/dwell
Benguerir (B02 washed-product yard)	stockpiles	2 × 191,512 t (total 383,024 t) · combined autonomy ≈6.1 d nom / ≈5.1 d design (BoD 5/6 d)	Q80130-BEN-B02-ME-BOD-0001 Rev B, IFR, 05-Feb-2025 §2.3.3	Confirmed	H	mine-yard model	reconcile earlier ≈10-day datasheet claim (document conflict)
Benguerir (cfg B)	3 qualities on 2 piles	in-pile segregation via SMS 1-m tracking (assumed)	—	Inferred	L	blending/inventory model	confirm operating rule (zones / campaigns)
Benguerir (B02)	machines	ONE twin-boom stacker `ST3**` 2100/2520 · reclaimers `RC2**` / `RC3**` 2600/3120 [TPH nom/design]	Q80130-BEN-B02-ME-BOD-0001 Rev B, IFR, 05-Feb-2025 §2.3.2 + project-team current-state instruction 21-Jul-2026	**Current configuration confirmed; `**` is source notation only**	H	current machine set	—
Benguerir (B02)	outlets	C2: washed 2600/3120 + **conditional raw** (hopper mod / RB9-RB10) · train: raw + washed 1000/1200 each (one station — sharing ASSUMED/TBC)	Q80130-BEN-B02-ME-BOD-0001 Rev B, IFR, 05-Feb-2025 §2.6	Confirmed; **train planning scope TBC**	H	route model — raw train + conditional raw→C2 may share **upstream raw-yard reclaim**; washed train + washed→C2 may share the **downstream product-yard reclaimers**; one shared loading station **ASSUMED/TBC**	confirm train planning scope + raw-route prerequisites
Benguerir (B01 2023 screening — connected upstream stage)	lines & storage	TM1 + TSM2 + TSM3 → common transfer tower `04BG-DT-01`; 1 reclaimer `04BG-RC-01`; stacker 2200 TPH nom; direct beneficiation bypass (mode deferred)	REP-0002 Rev B, 26-Jul-2023 (IFR)	Confirmed (PFDs indicative; vendor simulation pending)	M	connected upstream stage — tag mapping to B02 unproven, registers kept separate	reconcile B01↔B02 via PFD-0010/0011 + DBD-0001
Benguerir (Table 20 — upstream screened-product ore storage)	topology, machines & starting state	2 stockpiles × 200,000 t; 100,000 t assumed t0 each; 2 dedicated stackers plus 1 shared reclaimer; physical/as-handled rates 2,100/2,500 t/h	User-supplied Table 20 + project-team topology and starting-state instructions	Topology/capacity USER_CONFIRMED; t0 is an assumption rather than timestamped telemetry	H/M	**BG-MINE-1/2 are solver-backed**; source package still marks the reclaimer out of its scope, so operational authority remains TBC	obtain timestamped SMS stock/grade, source document revision, tags and GA
Benguerir upstream raw piles	shared raw grades + assumed t0 composition	`RAW-BT`, `RAW-BT-HIGH-CD` and `RAW-BT-LOW-CD` are mine-neutral. Demonstration t0: BG-RAW-1 = 50% BT + 50% High-Cd; BG-RAW-2 = 50% BT + 50% Low-Cd, 100,000 t per pile.	Louta BoD chemistry proxy + project-team scenario assumptions	Grade identities configured; composition and chemistry are ASSUMED, not live Benguerir assays	M	**BG-RAW-1/2, their stackers/shared reclaimer, inventories and jobs are solver-backed**; quality participates where a profile exists	replace with SQMS segments, dated assays and approved grade/receiving intervals
Louta 102MG mine storage	assumed demonstration baseline	`102MG-SP-01/-02`: **100,000 t assumed capacity each; 50,000 t t0 each (50%)**. SP-01 = 25,000 t BT + 25,000 t High-Cd; SP-02 = 25,000 t BT + 25,000 t Low-Cd. Components occupy schematic R0–R250 and R250–R500 ranges.	Project-team demonstration baseline, 26-Jul-2026; chemistry vectors from Louta BoD Rev B §6.1	**ASSUMED display values.** Actual capacity, live mass, grade order and route/rate/reach remain TBC. The documented **494,274 t belongs to 110/120LG**. Neither Louta PDF contains `102MG` or a 102MG→110LG route.	M	**reference only**; absent from solver inventory/machines/jobs/KPIs; `MODELLING_GAP[MISSING_ROUTE_RATE_REACH]`	capacity schedule, timestamped SMS/SQMS segments, surveyed repères and complete route/rate/reach matrix
Louta 110/120LG homogenized ore storage	topology, machines, capacity + starting state	**2 homogenized stockpiles** · shared `110LG-ST-01` 2,100 TPH wet · `120LG-RC-01/-02` 3,000 TPH wet · common RB4/RB5 to L1/L2 · total **494,274 t → 2 × 247,137 t DERIVED** · starting and exact S0 terminal mass **123,568.5 t each ASSUMED** · displayed OL1/OL2 bands ASSUMED	Process description Rev B §3.1 pp.7–8; total/profiles: Louta BoD Rev B. **Document-control inconsistency:** cover `REP-00002` vs body `REP-00004`.	Counts/topology/rates CONFIRMED in the tagged description; split, stocks, bands and pairing ASSUMED/DERIVED. **Count conflict:** BoD p.20 says one ore reclaimer; tagged	M	`110LG-OL1/-OL2`, `110LG-ST-01` and `120LG-RC-01/-02` are **solver-backed**; bands remain display metadata	live stocks/assays, pile schedule, reach/GA, common-route capacity/concurrency and approved campaigns

Site / domain	Parameter / rule	Value [unit]	Source · rev · ref	Status	Conf.	Design impact	Owner / action
				description says two; tagged configuration used.			
Louta washing / beneficiation	line configuration, cases, recovery + residence	**2 parallel L1/L2 lines, 3 Mt tsm/year each**; BT **584 t/h / 68.22%**, High-Cd **579 / 68.83%**, Low-Cd **588 / 67.71%**; scrubber **2.5–3 min CONFIRMED**; overall line **30 min ASSUMED**	Process description Rev B p.5; case data and scrubber: Louta BoD Rev B; overall residence: project-team instruction 26-Jul-2026	Case data and scrubber residence CONFIRMED design values; 30 min and S0 BT→L1→TBT / Low-Cd→L2→BT mapping **ASSUMED**. PDFs do not map cases to TBT/BT/MT.	M	**solver-backed process-transfer stages** with mass yield, no same-bucket passthrough, cumulative matured-mass availability and terminal WIP = 0 t; not `Machine` objects	validate residence; obtain availability, approved campaign map and shared-tank/filter concurrency rules
Louta 130/140LG product handling	stack/reclaim topology + S0 envelope	One `130LG-ST-01` on SL11; `140LG-RC-01/-02` at 3,000 TPH; RL1/RL2 → flip gates → RL3/RL4 → outbound `BL` . S0 uses a **1,000 TPH ASSUMED stacker envelope** and conservative pairing/confluence exclusivity.	Louta Process description Rev B §3.1 p.8; S0 envelope/pairings: project-team assumptions	Tagged machine configuration CONFIRMED; rate/pairing/confluence TBC. **Count conflict:** BoD p.20 says one product reclaimer; tagged description says two. Neither PDF labels the outbound as C1.	M	`130LG-ST-01` , `140LG-RC-01/-02` , LT-P1/P2 and C1 material movement are **solver-backed**. The operator schedule labels 140LG work by its physical **reclaim** operation; the former green dispatch category is reserved for explicit non-material positioning.	product arrangement, stacker/SL11 rating, reach, confluence capacity and live state
Louta S0 stocked-route policy	explicit purpose targets and source boundary	Exact: C1 **8,000 t**; recovered L1/L2 **4,000 t each**; terminal OL1/OL2 **123,568.5 t each**; terminal LT-P1/P2 **95,756 t each**; WIP **0 t**. Two **10,000 t ASSUMED mine-handoff** sources feed 110LG and are not 102MG. Coverage reward 0; flow penalty 1.	Project-team demonstration policy, 28-Jul-2026; implementation configuration	**ASSUMED policy**, not a source-document operating requirement. Opening product stock can otherwise satisfy demand with legitimate upstream idleness; direct ore/product bypasses are documented.	M	every S0 job has a corridor, process-output or terminal-stock purpose; no coverage/token jobs; switch-off returns demand-only mode	replace with approved source allocation, end-stock, bypass and campaign policy
Louta machine disposition	selected solver machines versus gaps	Structured result for `110LG-ST-01` , `120LG-RC-01/-02` , `130LG-ST-01` , `140LG-RC-01/-02` : scheduled, unavailable, no inventory, no target/free capacity, missing route, shared-route peer selected, or target met without machine. PDFs do not call either pair duty/standby.	Solver result semantics + Louta source review	Model behaviour; disposition is explanatory, not evidence of operating duty	M	explains justified work or legitimate idleness without manufacturing activity	connect live availability, routes, inventories and operating duty rules
Louta L02 product piles	starting inventory + grade distribution	Capacity **191,512 t each**; starting mass **95,756 t each (50%)**. **LT-P1 = 60% TBT + 40% BT** → BPL 71.6% · Cd 25.2 ppm · MgO 0.92% · SiO_2 2.32% · moisture 12%. **LT-P2 = 60% BT + 40% MT** → BPL 66.0% · Cd 32.0 ppm · MgO 1.22% · SiO_2 3.08% · moisture 12%. Yard aggregate = 30% TBT / 50% BT / 20% MT.	Capacity derived as equal split of documented 383,024 t total; starting state and distribution: project-team demonstration instruction 24-Jul-2026	**ASSUMED distribution.** TBT/BT/MT are washed demonstration vectors; Louta operating mapping/cut-offs TBC. Bands are mass shares, not spatial order. The shared raw `RAW-BT-*` grades stay a separate material state and are never applied to these washed-product lines.	M	solver uses each pile's mass-weighted aggregate `StockSpace.quality` ; no profile-selective reclaim or zone allocation	replace with timestamped pile/segment masses, surveyed boundaries, active reclaim face and Grade/SQP mapping
Louta L02	combined product specification	BPL 68.29% · MgO <0.6% · CO_2 6.81% · SiO_2 <4% · Cd 17 ppm · moisture <22% · SG 3	`Q80130-LOU-L02-PR-BOD-0001` Rev B, 12-Apr-2023, p. 18	Documented design specification; retained under the source's own stage wording, **not relabelled as washed-only**	H	reference only; does not replace TBT/BT/MT demonstration vectors	confirm operating product grade/SQP IDs, acceptance limits and sampling basis
Benguerir Table 20 ↔ B02	reclaimer capacity conflict	Table 20: 2,100/2,500 TPH humid vs B02 expansion interface: 2,600/3,120 TPH — whether B02 upgrades, replaces or supplements the Table 20 reclaimer is TBC	Table 20 vs Q80130-BEN-B02-ME-BOD-0001 Rev B §2.3.2	**Conflicting — not resolved by inference**	M	raw→C2 demonstration at 2,600 TPH assumes the B02 interface selected and the Table 20 bottleneck resolved	equipment list / PFD proving equivalence or supersession
Benguerir (cfg A)	stream rates	line feed 1300 · primary undersize 1208 · secondary feeds 2 × 604 · crusher 541 · oversize reject 89 · product handling 1055 [TPH]	REP-0002 Rev B	Confirmed in doc	M	none yet (out of model)	—
Benguerir (cfg A)	reject total	541 + 89 = **630** vs stated **628** TPH	REP-0002 Rev B	**Conflicting**	—	preserved, not normalised	resolve with vendor
Benguerir (cfg A)	undersize vs handling	**1208** TPH screen undersize vs **1055** TPH product-handling capacity	REP-0002 Rev B	**Conflicting / unexplained**	—	possible surge / split logic unknown	resolve flow basis
Benguerir B01↔B02	process relationship	raw/screened mine storage → washing/filtration/settling → washed-product storage → C2/train (+conditional raw→C2 · raw/washed→train)	Q80130-BEN-B02-ME-BOD-0001 Rev B, IFR, 05-Feb-2025 §2.6	**Confirmed at stage level; exact phase/tag/PFD reconciliation TBC**	H	staged Benguerir model	obtain PFD-0010/0011 + BFD DBD-0001 (tag mapping)
Benguerir washing lines	planning lifecycle and rate controls	`BEN-WASH-L1/L2/L3` are executable recovery-bearing stages with scenario-configurable state and physical rate	Project-team model baseline through 8-Sep-2026; operating/as-built status not independently evidenced	Implemented planning resources; live status remains TBC	M	all three can create feed/recovery/stacking work and obey whole-run or timed outage/derate controls	obtain current commissioning status, canonical tags, capacities and effective dates
Benguerir washing / beneficiation	yield, moisture and residence	3 configurable washing campaigns; scrubber 3 min CONFIRMED; overall line 30 min ASSUMED; editable grade yield/raw moisture/wet-product moisture	`Q80130-BEN-B02-PR-BOD-0001` Rev 0 + project-team conversion/residence assumptions	Mixed evidence; conversion defaults are visible scenario assumptions	M	**solver-backed recovery-bearing process transfers**; yield reduces dry product mass and residence delays product availability	validate campaign mapping, yield, moisture and overall residence from operating data
Louta	stockpiles	`MINE STOCKPILE-01` / `-02` (colored base arrangement)	cropped layout drawing (doc no./rev/title block TBC)	Confirmed visible	M	2 piles exist; nothing quantitative derived	obtain full drawing + legend
Louta	equipment tags	`102MG-ST-01` · `102MG-TR-06` · `CP-07` · `CP-08` · `CP-10` · `CP-11`	cropped layout drawing	Confirmed visible; functions/connectivity TBC	M	no capacities/machine counts/routes derived from the crop	machine-to-pile / route matrix
Louta	`TO DIGUE (OPTION-2)` grey layout	separate option incl. RAW/PRODUCT STOCKPILE BENEFICIATION-01/-02	cropped layout drawing	Option — **excluded from Phase-1 baseline**	—	not in model	confirm selected option
MPH SMS · ASP	route semantics	`AspRoute` = (Device, JobType, free-text Location) only — no network model	FRS §3.1	Confirmed	H	layer owns route graph (§14.2)	ICD
MPH SMS · ASP	rule severities	5 ERROR · 2 WARNING · 1 INFO (§15.7)	FRS UC-22…30	Confirmed	H	validation-gate design; warning promotion	governance decision
MPH SMS · ASP	quality model	1 job = 1 grade; SQP outside ASP; Blending Simulation separate ("no role in ASP"); Sub-Grade behaviour unresolved	FRS Q&A	Confirmed / **Unresolved (SQP/Sub-Grade)**	M	blending owned by this layer	ICD / configuration question
MPH SMS · ASP	inventory cache	per-pile 1-m X segments (mass · prevalent grade · age), ~10-min or on-demand refresh	FRS	Confirmed	H	twin sync design	—
All stockpiles · SMS/ASP	longitudinal job location / repère	Operator scale **R0–R1000 = 1,000 equal intervals**; jobs use a range such as **R400–R600**. Physical X metres remain canonical and are calculated per pile.	Start/End positions and 1-m inventory: Innomotics FRS Table 26, UC-22/23/24/26/28/30 and §3.7.1; repère name/scale: project team 24-Jul-2026	**FRS position model Confirmed; repère convention User-confirmed; physical calibration TBC**	H/M	job authoring, grade placement, machine display, inventory selection and spatial checks	obtain each pile's R0 endpoint/direction, surveyed origin/usable length and production ICD StartX/StopX units/scale
Simulator mimic	Initial / Final state and planning controls	**Initial** = scenario t0; **Final** = terminal weekly inventory. Physical fill is proportional; hover exposes component mass/quality. Louta 102MG remains reference-only; the six Benguerir piles and Louta 110LG/product spaces are solver-backed.	Project-team UI instructions and implementation through 8-Sep-2026	Implemented engineering behaviour; scenario values are not live telemetry	M	shared Monitor/Routing diagram, state/rate controls and terminal-plan inspection	validate against live SMS/SQMS and surveyed geometry

Site / domain	Parameter / rule	Value [unit]	Source · rev · ref	Status	Conf.	Design impact	Owner / action
Interfaces	RabbitMQ telegram schemas	`AspPlanItem` / `AspEvaluatePlanResponse` / `JobUpdated` / `SpaceValueRequest/Response` — ICD not yet received	FRS (hypothesis H1)	TBC	—	integration design	request ICD (vendor workshop)
Economics	GradeCostModel curves	PAP specific consumptions vs feed grade + rock values per quality	—	**Required, not received**	—	w_3/w_4 objective terms	process engineering / finance

17. Simulation Readiness

What the allocation/simulation model needs, by readiness class:

Class	Items
Implemented engineering MVP	deterministic weekly CP-SAT planning; dated multi-campaign demand; physical inventory/rate constraints; washing yield/moisture conversion; editable grade catalogue and receiving intervals; timed equipment/conveyor/line controls; global status diagram; results/KPI and day-filtered job schedule; M16 fixed/optimized pile-construction blending.
Configurable assumptions (visible and editable)	travel/setup/minimum separation · weekly demand campaigns · initial inventory and pile distributions · yield/moisture profiles · chemistry vectors and receiving intervals · work calendars · equipment/routing states and physical rate caps · objective weights. Demonstration B02 defaults are MT 75.73%, BT 66.63%, TBT 62.64%, raw moisture 12% and wet-product moisture 17.5%; these are assumptions, not live measurements.
Required before a realistic operational simulation	timestamped SMS/SQMS stocks and assays · validated grade-class intervals and raw↔washed mappings · per-pile repère calibration and metric GA · route confluences/capacities, reach and transition data · current machine/conveyor/plant state and calendars · validated washing residence/yield/moisture by campaign · drying intake · approved operating contingencies and a real GradeCostModel. More detailed within-pile layer/zone reclaim composition and the two non-M16 blend points are not implemented.
Required only before live SMS integration	RabbitMQ ICD / telegram schemas + credentials (H1), including confirmed StartX/StopX units, axis origin/orientation and scale · site-specific mine-SMS configuration (H2 **resolved at solution level — identical Innomotics/YardMaster SMS at all three sites, user-confirmed**; per-site endpoints, queue/exchange names, Device/Pile IDs, Material/Grade/SubGrade mappings, AspRoutes and permissions remain TBC) · plan co-authoring governance & write access (H3) · final configured Material/Grade/SQP/Sub-Grade behaviour · LIMS/CMS/ERP endpoints · OT/IT & cybersecurity engineering

Verification boundary (8 September 2026). The focused blending/integration suite reports **61 passing tests**. A real seven-day default solve returned FEASIBLE with exact 12,000 t blend construction and no reclaim before the 48-hour boundary. The broader repository run reported **755 passes and 55 legacy/baseline failures**; therefore this report does not claim that every historical UI/report assertion is green.

18. Acceptance Criteria (measurable)

Numeric targets marked TBC are to be fixed at Phase-1 design review; the *measures* are fixed now so every criterion is testable in simulation before live integration.

#	Criterion	Measure	Target
1	No impossible route allocations	jobs dispatched whose path is absent from the route graph's valid-path set	**0**
2	No machine double booking	overlapping jobs per device (ASP Same-Device-Overlap ERRORs on AUTO items)	**0**
3	Inventory conservation	per-space physical mass-balance closure over the full 168-hour week, including washing conversion and carried WIP	≤ ε % drift (TBC)
4	Destination quality compliance	% of buckets with delivered mass-weighted quality within spec (eq. 3), per destination	≥ target % (TBC)
5	Demand service	`Σ delivered / Σ demand` per destination; `unmet` tracked per bucket	≥ target % (TBC)
6	Safety-stock compliance	% of horizon with `inventory ≥ safety level`, per space class	≥ target % (TBC)
7	Solve time	tactical optimiser wall-clock per rolling solve	≤ limit (TBC, order of minutes)
8	Replanning latency	event (JobUpdated deviation / fault) → new dispatched plan	≤ limit (TBC, order of seconds–minutes)
9	ASP validation success	AUTO plan items passing ASP with **zero ERRORs** at first submission	≥ target % (TBC)
10	Operator override preservation	human-authored/modified items never edited or reverted by the layer	**100 %**
11	Deterministic replay	re-running a stored scenario (inputs + config + seed) reproduces the identical plan	**exact match**
12	Multi-campaign daily demand	all same-unit/day campaign quantities, windows and grades survive input→solve→result reconciliation	**exact conservation; no positive-window overlap**
13	Timed contingency response	outage/derate at a non-grid minute changes no work before the boundary and produces no use above availability/rate after it	**0 violations**
14	Blend construction integrity	scheduled grade tonnes equal selected matrix, total equals batch, all bounds/quality intervals hold and no reclaim starts before exact readiness	**exact integer-tonne closure; 0 early reclaim**
15	Stage mass traceability	physical raw, TSM and wet-product values reconcile with the selected yield/moisture profile	**within configured numeric tolerance**

Abbreviations

Abbrev.	Meaning	Abbrev.	Meaning
ACS	Anti-Collision System	ASP	Advanced Stockyard Planning
BPL	Bone Phosphate of Lime	CMS	Maintenance / Asset system
DCS	Distributed Control System	GA	General Arrangement
HMI	Human-Machine Interface	ICD	Interface Control Document
LIMS	Laboratory Information Management System	MES	Manufacturing Execution System
MPH	Mzinda Phosphate Hub	MTPY	Million tonnes per year
OPC UA/HDA	OPC Unified Architecture / Historical Data Access	PAP	Phosphoric Acid Plant
PFD	Process Flow Diagram	PLC	Programmable Logic Controller
PSD	Particle-Size Distribution	SMS/SQMS	Stockpile (Quality) Management System
SPH	Safi Platform Hub	TBE	Technical Bid Evaluation
TBC/TBD	To Be Confirmed / Determined	TPH	Tonnes per hour
TSM	Zero-moisture merchant dry mass		

Working draft for scope and functional-design review. Evidence classifications and unresolved actions remain authoritative in §16.